Fixed Point Iteration Schemes

One-Step vs Two-Step Schemes

One-Step (Picard):

$$\boxed{x_{n+1} = Tx_n}$$

Two-Step (Ishikawa):

$$\boxed{y_n = (1 - \alpha_n)x_n + \alpha_n T x_n}$$

$$\boxed{x_{n+1} = (1 - \beta_n)x_n + \beta_n T y_n}$$

Convergence Characteristics

Picard:

- Faster convergence
- Works for contractions
- Limited to certain mappings

Ishikawa:

- More flexible
- Works for nonexpansive maps
- Controlled by parameters